

Bora Uner

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EDUCATION

Michigan State University

Bachelor of Science in Computer Science, Additional Major in Economics; GPA: 3.93/4.0

- Honors College; Dean's List (all semesters); Certification: Algolia Certified Developer (ACD)

East Lansing, MI

Expected May 2027

EXPERIENCE

Full-Stack Software Engineering Intern

May 2026 – Aug 2026

Corewell Health

Grand Rapids, MI

- Strengthened corewellhealth.org disaster recovery by replicating its API Gateway stack to a secondary AWS region (us-east-2) with AWS CDK and adding Route 53 health-check routing for automatic failover from the primary region (us-east-1).
- Replaced legacy breadcrumb navigation with regional location badges across all location pages using TypeScript and SCSS, partnering with the design team to resolve layout defects in hero components and improve wayfinding consistency sitewide.
- Built a dynamic “Related Pages” component that maps each location page’s taxonomy terms to a service-line dictionary, surfacing relevant service links across all location pages to improve content discoverability.
- Enhanced the site footer with a new notice section and worked with the design team to fix visual defects in production.
- Delivered the intern group’s final presentation to 500+ employees, including the CTO, and personally answered the CTO’s follow-up questions; separately demoed the regional badges feature to 100+ Digital Services staff and directors.

Undergraduate Learning Assistant, CSE 380

Aug 2026 – Present

Cloud Computing & Information Management, MSU Department of Computer Science & Engineering

East Lansing, MI

- Host help-room sessions guiding students through course projects that build Flask REST APIs, containerize microservices with Docker, and implement JWT-based authentication and authorization.
- Debug student code one-on-one and explain the cloud and information-management concepts behind each project.

Resident Assistant

Aug 2025 – Present

MSU Student Life & Engagement

East Lansing, MI

- Foster an inclusive community for 48 residents through academic programming, conflict resolution, and peer mentoring.

Undergraduate Learning Assistant, Physics

Jan 2024 – Jan 2025

MSU College of Natural Science

East Lansing, MI

- Led weekly student groups applying physics to real-world problems and delivered tailored feedback on their work.

PROJECTS

Object-Oriented Programming Projects | C++, wxWidgets, UML

Jan 2025 – Aug 2025

- Developed two C++ desktop applications, a Mastermind-style puzzle game and an animated machine simulation, applying the Factory, Visitor, Adapter, Observer, Composite, and Iterator design patterns.
- Built interactive GUIs with wxWidgets, including event handling, custom rendering, and keyframe animation.
- Led agile sprints for a 7-member team, establishing UML models and coding standards to coordinate development.

High-Performance Bit Matrix Rotation | AVX2 Intrinsics, Cache Optimization

Jan 2025 – Apr 2025

- Engineered a cache-blocked rotation algorithm for large bit matrices using loop unrolling and prefetching.
- Vectorized block transformations with AVX2 intrinsics to accelerate bit manipulation and reduce memory overhead.
- Raised performance from tier 4 to tier 94 of 100 on the course benchmark, where each tier enlarges the matrix and tightens the time cap, with correct handling of both even and odd matrix sizes.

Parallel Ray Tracing & N-Body Physics Simulation | C++, Cilk

Jan 2025 – Aug 2025

- Built a parallel ray tracer with ray-sphere intersection and Lambertian shading using Cilk parallel loops; optimized rendering with projection-based bounding volumes, precomputed lighting, and a custom hybrid quicksort.
- Implemented an N-body physics simulation with collision handling, resolving race conditions to scale across multiple cores.

Social Media Backend | Python, Flask, SQLite, Docker

Aug 2024 – Jan 2025

- Built a REST backend with secure authentication, a follower graph, and engagement features on an indexed SQLite schema.
- Refactored the monolith into Dockerized microservices to separate concerns and simplify deployment.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, C, C++, SQL

Web: HTML, CSS, SCSS, React, Gatsby, GraphQL, REST APIs, Contentstack (Headless CMS)

Cloud & Backend: AWS (CDK, Lambda, API Gateway, S3, CloudFront, Route 53), Infrastructure as Code (IaC), Flask, Microservices, JWT Authentication, Algolia

Tools & Practices: Docker, Git, Linux, Jira, CI/CD, Agile/Scrum, Object-Oriented Programming (OOP), Design Patterns, UML, System Design, Unit Testing